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Stability study for preparation of haematology samples

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Conclusion

- **For all parameters, stability is assured as long as the samples are stored at 2-8°C for two days. For all parameters except platelet count, this period can be extended up to three days.**
- **Except for platelet count, there is no evidence that adding a fixative prolongs stability. Adding a fixative however affects the assigned value of several parameters, which may cause a problem of commutability.**

Using fresh blood samples for the complete blood count EQA program raises the challenge of guaranteeing stability. A study was set up to compare three different protocols for preserving fresh blood samples for haematology parameters. Two questions were of interest: (1) which protocol is the best for preserving samples and (2) how many days can the samples be stored before analysis?

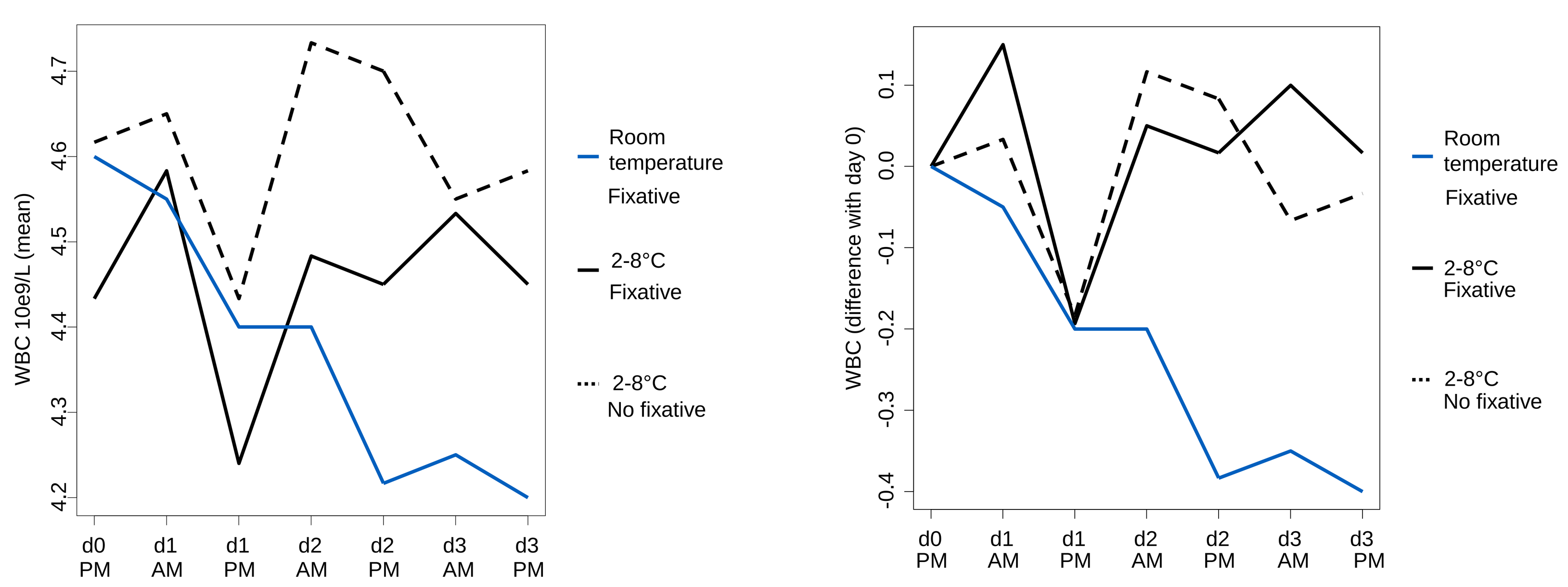
Methods

- One blood donation from one single donor was used.
- The protocols were:
  - **(1)** Fixative (glutardialdehyde 0.025%) + room temperature
  - **(2)** Fixative + 2-8°C
  - **(3)** No fixative + 2-8°C
- The samples were analysed at day 0 PM, day 1 AM, day 1 PM, day 2 AM, day 2 PM, day 3 AM and day 3 PM.
- The evaluated parameters were: white blood cell count (WBC), red blood cell count (RBC), haemoglobin (Hb), haematocrit (Hct), mean corpuscular volume (MCV) and platelet count.
- The statistical analysis was performed by equivalence testing by means of confidence intervals around the mean changes based on the results of an Analysis of Variance after exclusion of outliers. For each protocol, comparisons with day 0 pm were made.
- The analyser used was ABX Micros ES60 (HORIBA Medical).

Results

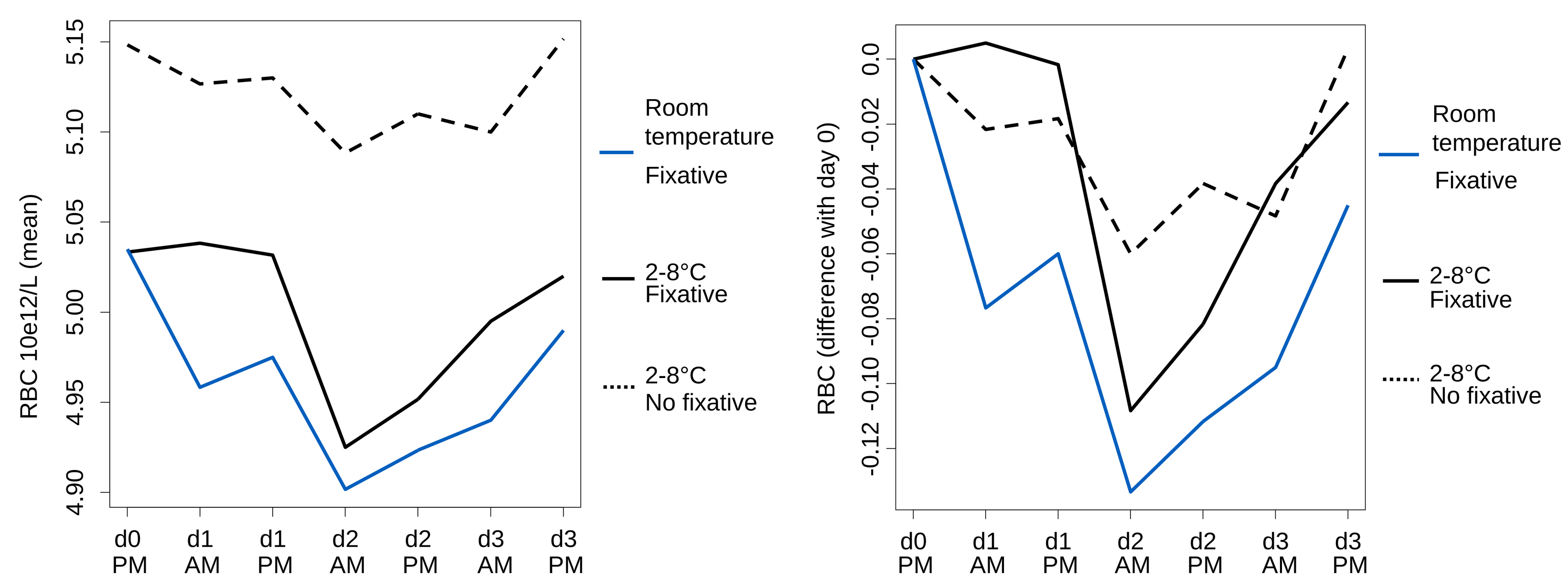
Data are displayed graphically by showing the mean concentration for each protocol at each time and the difference of the means for a certain time with the measurement at day 0 pm.

WBC



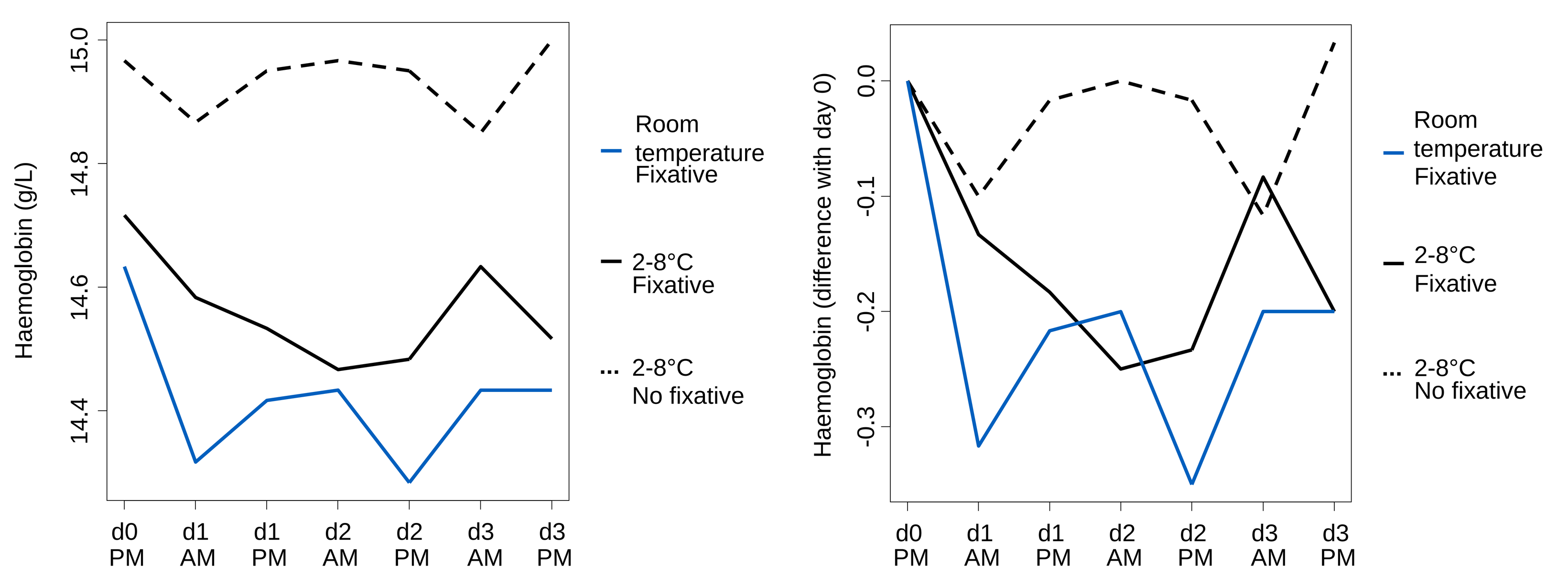
➔ *There is lack of evidence that a sample stored at room temperature is still stable at day 1 in the afternoon.*
➔ *Samples stored at 2-8°C stay stable up to day 3 in the afternoon.*
➔ *The fixative does not seem to have any effect on the stability for the period considered in this study.*

RBC



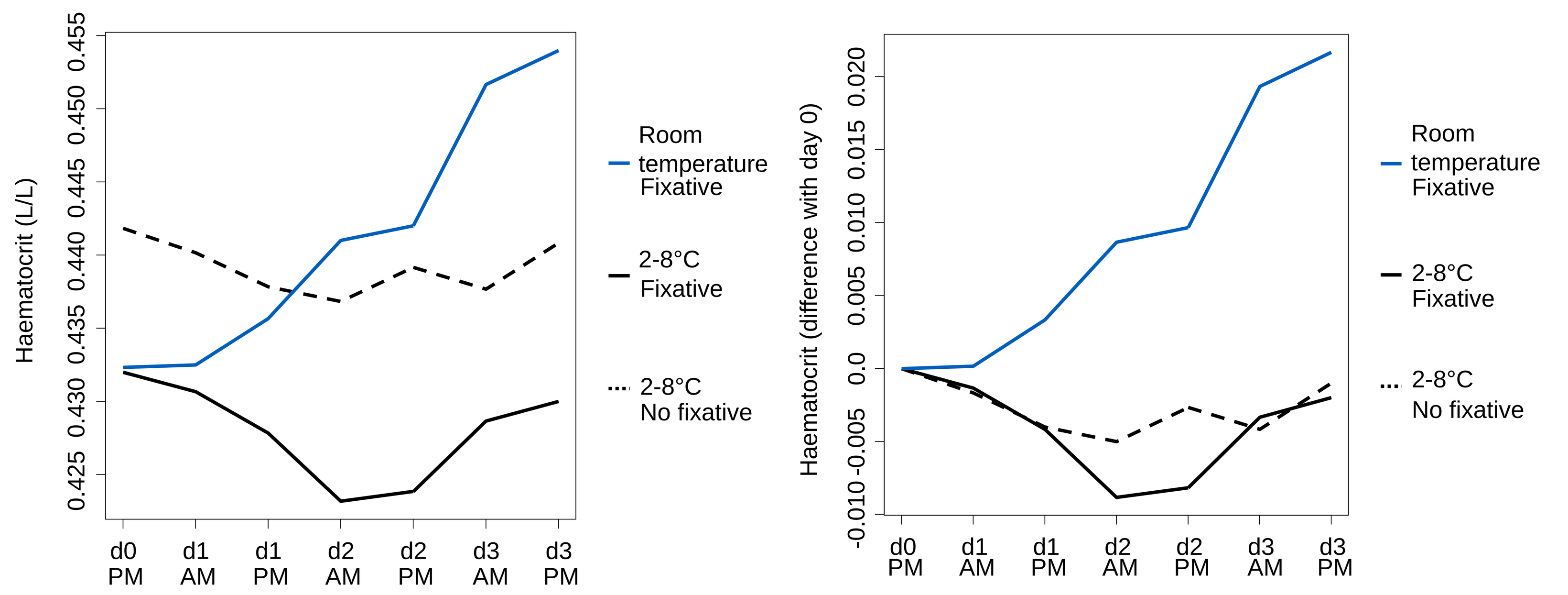
➔ *A large discrepancy can be observed between the samples that contained the fixative and the samples that did not contain any fixative.*
➔ *The storage on room temperature makes the RBC to drop.*

Haemoglobin



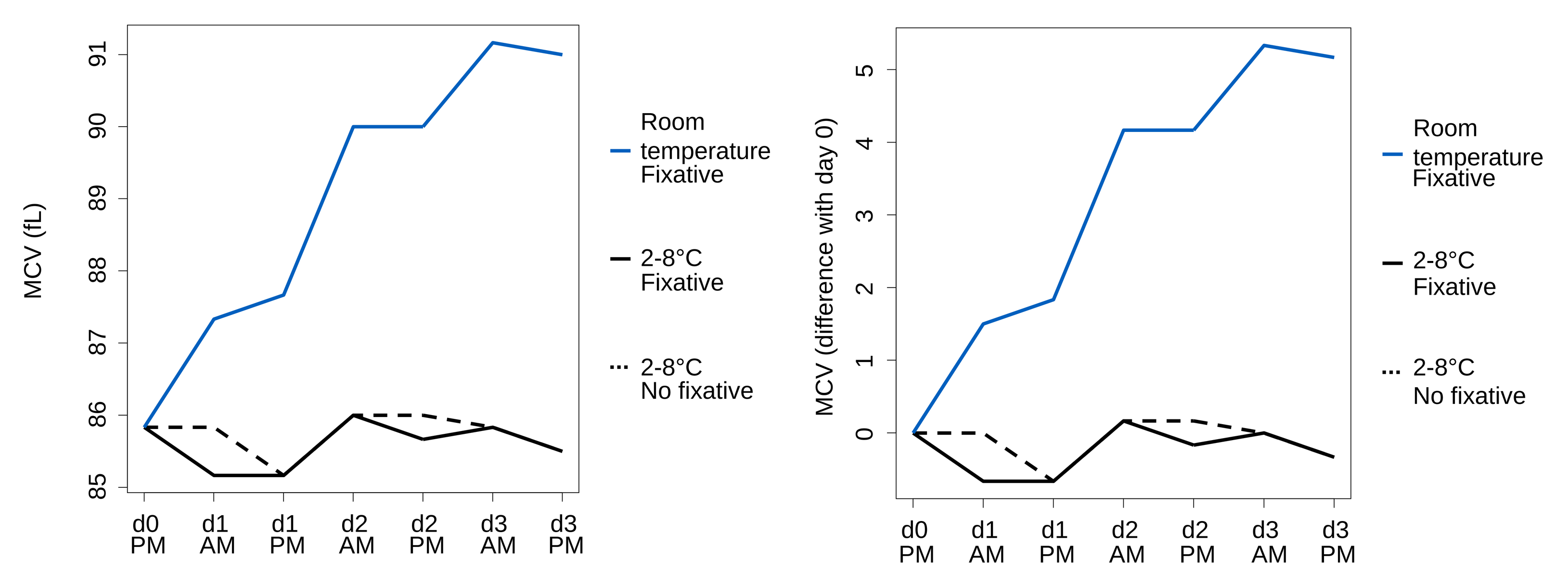
➔ *Stability is assured regardless of the protocol.*

Haematocrit



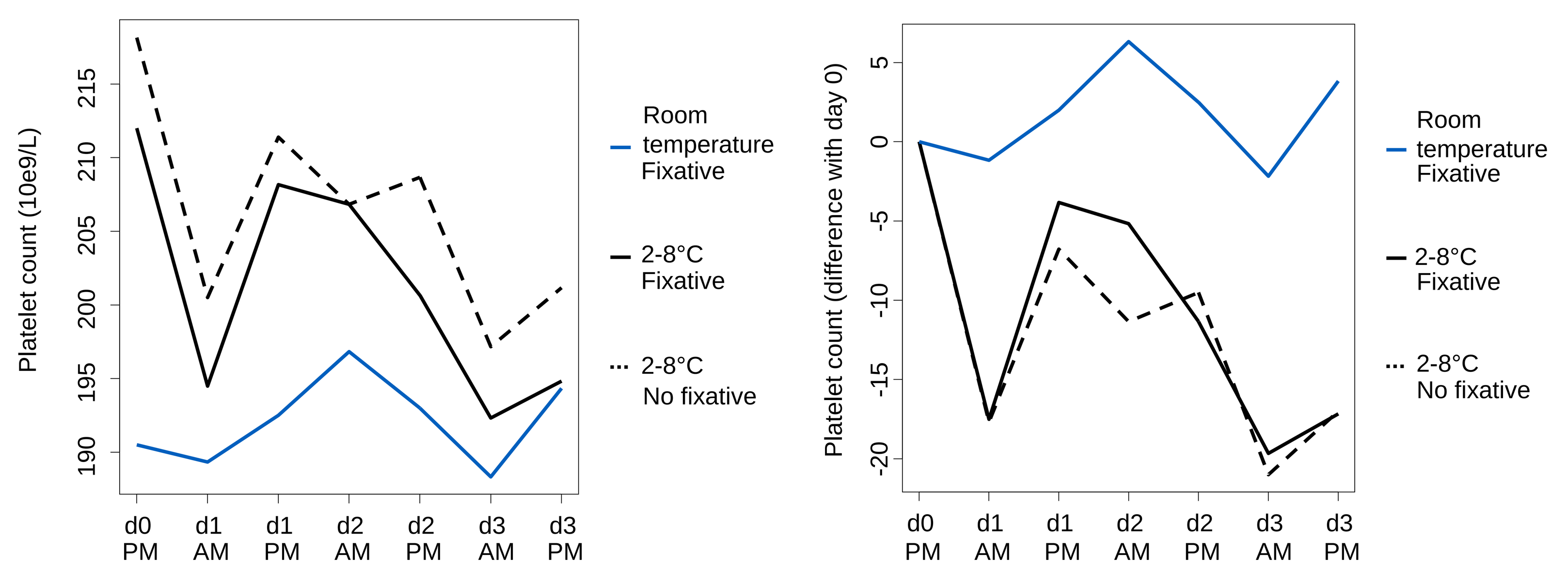
➔ *Storage on room temperature assures stability only for the first day, storage at 2-8°C assures stability till day 3, afternoon.*

MCV



➔ *MCV appears to be very stable when the sample is kept at 2-8°C, but stability is not assured when the sample is kept at room temperature. Adding the fixative does not have an observable effect.*

Platelet count



➔ *Adding a fixative appears to have an effect on the measured platelet count.*
➔ *Storing the sample at room temperature appears to be the best choice; storing at 2-8°C assures stability until day 2.*

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